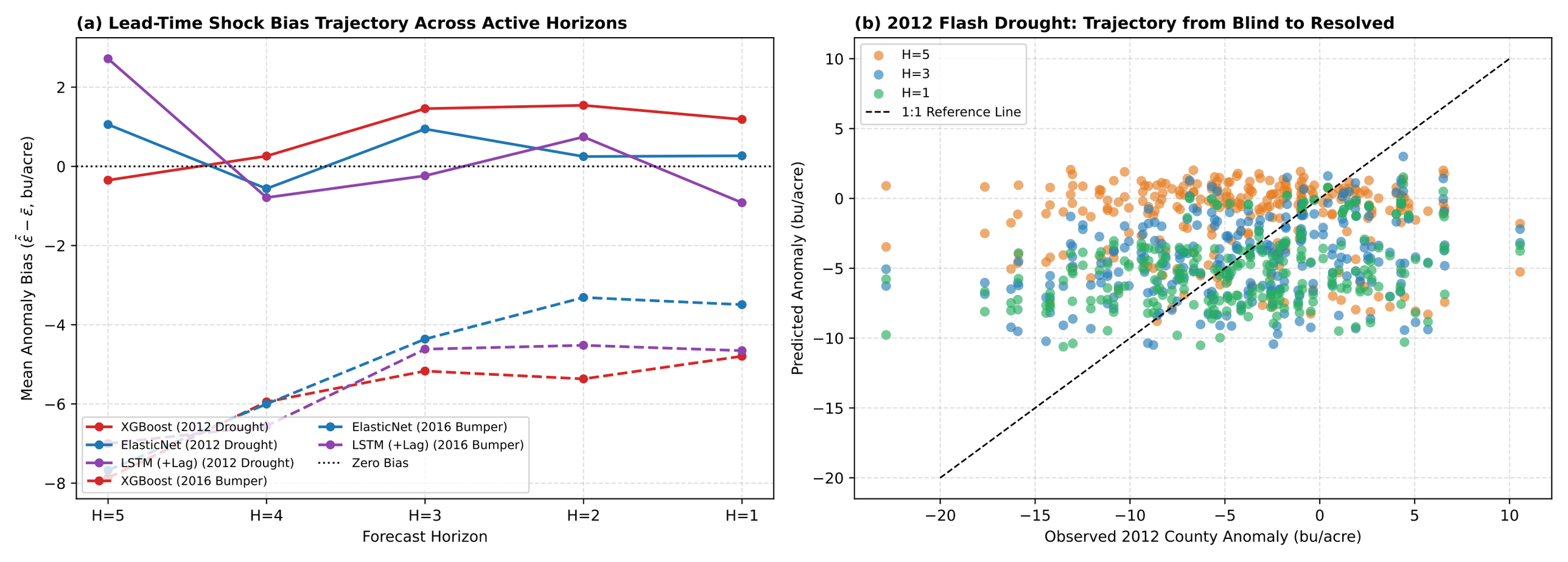
(a) Lead-Time Shock Bias Trajectory Across Active Horizons
Mean Anomaly Bias ($\bar{\hat{\varepsilon}} - \bar{\varepsilon}$, bu/acre)
Forecast Horizon
H=5
H=4
H=3
H=2
H=1
2
0
−2
−4
−6
−8
XGBoost (2012 Drought)
ElasticNet (2012 Drought)
LSTM (+Lag) (2012 Drought)
XGBoost (2016 Bumper)
ElasticNet (2016 Bumper)
LSTM (+Lag) (2016 Bumper)
Zero Bias
(b) 2012 Flash Drought: Trajectory from Blind to Resolved
Predicted Anomaly (bu/acre)
Observed 2012 County Anomaly (bu/acre)
H=5
H=3
H=1
1:1 Reference Line
10
5
0
−5
−10
−15
−20